Theory : Threat LandscapeTypes of cyber threats: Phishing, ransomware, insider threats, etc.:Simulate a phishing attack in a controlled environment using open-source tools (e.g., SET toolkit).Demonstrate email spoofing and prevention techniques.

Tasks :-

1.Create a table listing 3 real-world examples each of phishing, ransomware, and insider threats that could affect apps like WhatsApp, Flipkart, or Paytm. For each, describe in 1-2 lines how the attack works.

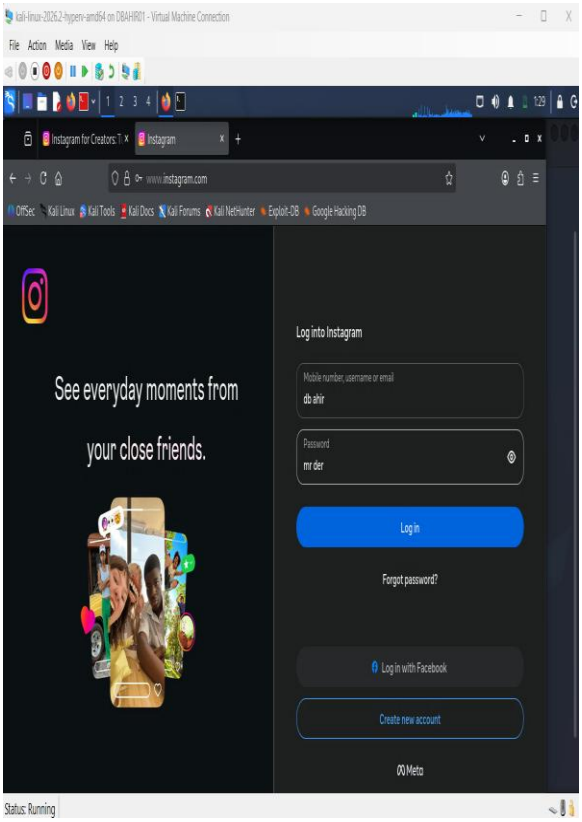
Ans :-

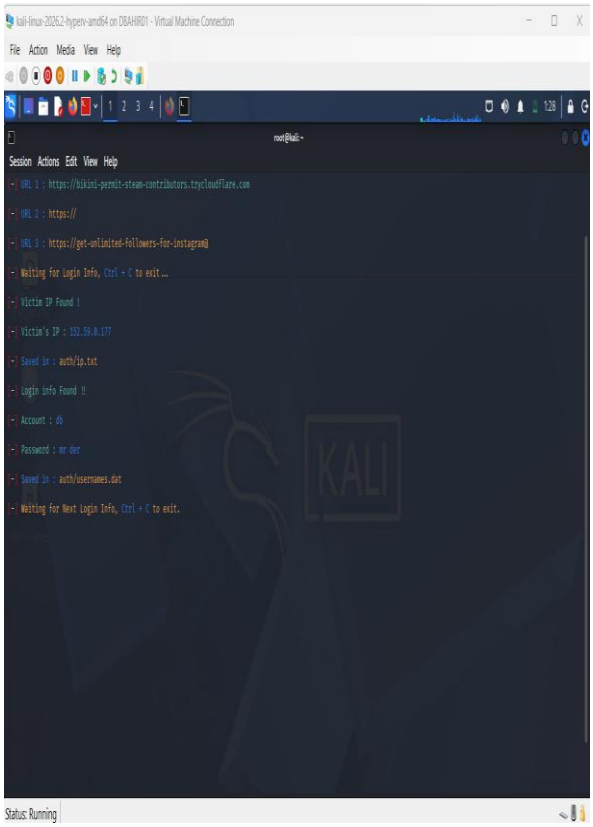
Threat Type	Real-World Example	How the Attack Works
Phishing	1. Fake WhatsApp Verification Link	Attackers send fake WhatsApp login links and trick users into sharing OTPs or verification codes to take over their accounts.
Phishing	2. Fake Flipkart Discount Offer	Attackers create fake shopping websites offering huge discounts to steal users' login credentials and payment details.
Phishing	3. Fake Paytm KYC Update	Users receive fraudulent SMS messages claiming their Paytm KYC has expired. Clicking the link may expose personal and banking information.
Ransomware	1. WhatsApp Support System Attack	Ransomware could infect a company's internal systems through a malicious attachment, encrypting customer support records and disrupting services.
Ransomware	2. Flipkart Warehouse System Attack	Attackers infect warehouse computers with ransomware, encrypting inventory and order management data and delaying deliveries.
Ransomware	3. Paytm Employee Computer Attack	Malware could encrypt internal employee files and business systems, disrupting operations until recovery from backups.
Insider Threat	1. WhatsApp Employee Data Misuse	An employee with authorized access could improperly view or disclose confidential user information.
Insider Threat	2. Flipkart Customer Data Theft	An employee could misuse database access to copy customer names, addresses, or order details and share them without authorization.
Insider Threat	3. Paytm Unauthorized Data Access	An employee with privileged access could misuse customer or transaction records for personal benefit or disclose them to outsiders.

**2.Use the Social Engineering Toolkit (SET) to simulate a basic phishing page that mimics a login screen for a popular app (e.g., Instagram or Zomato). Capture and display the entered credentials in your local environment.

Hint: Use the 'Website Attack Vectors' option in SET and select 'Credential Harvester Attack Method'.**

Ans :-

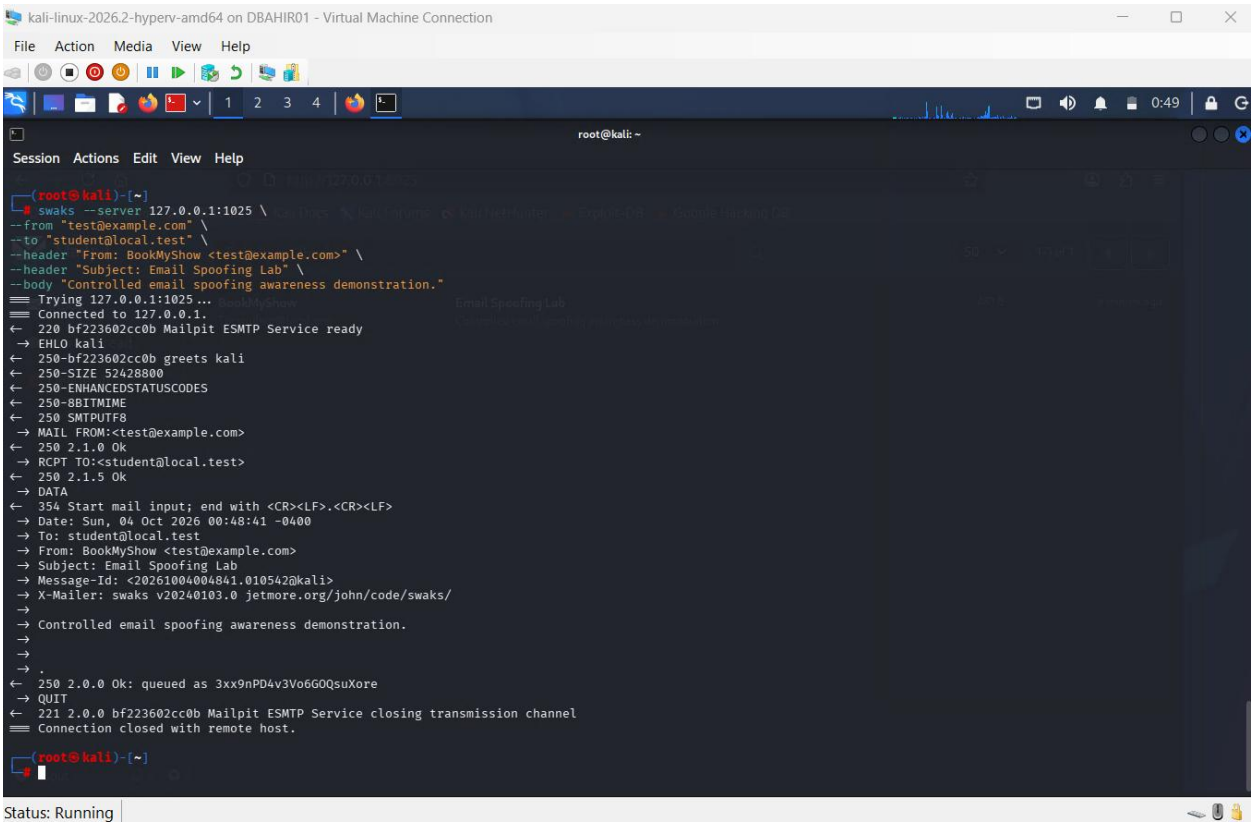


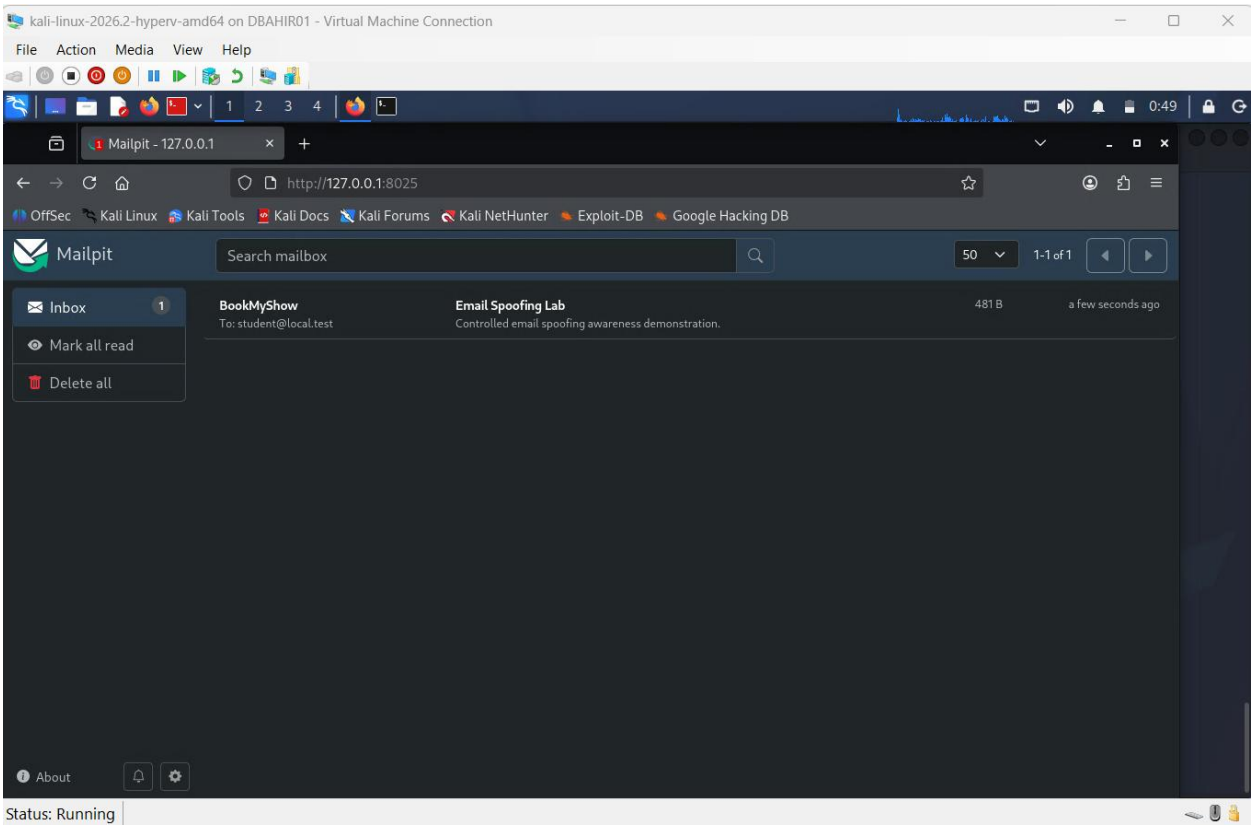


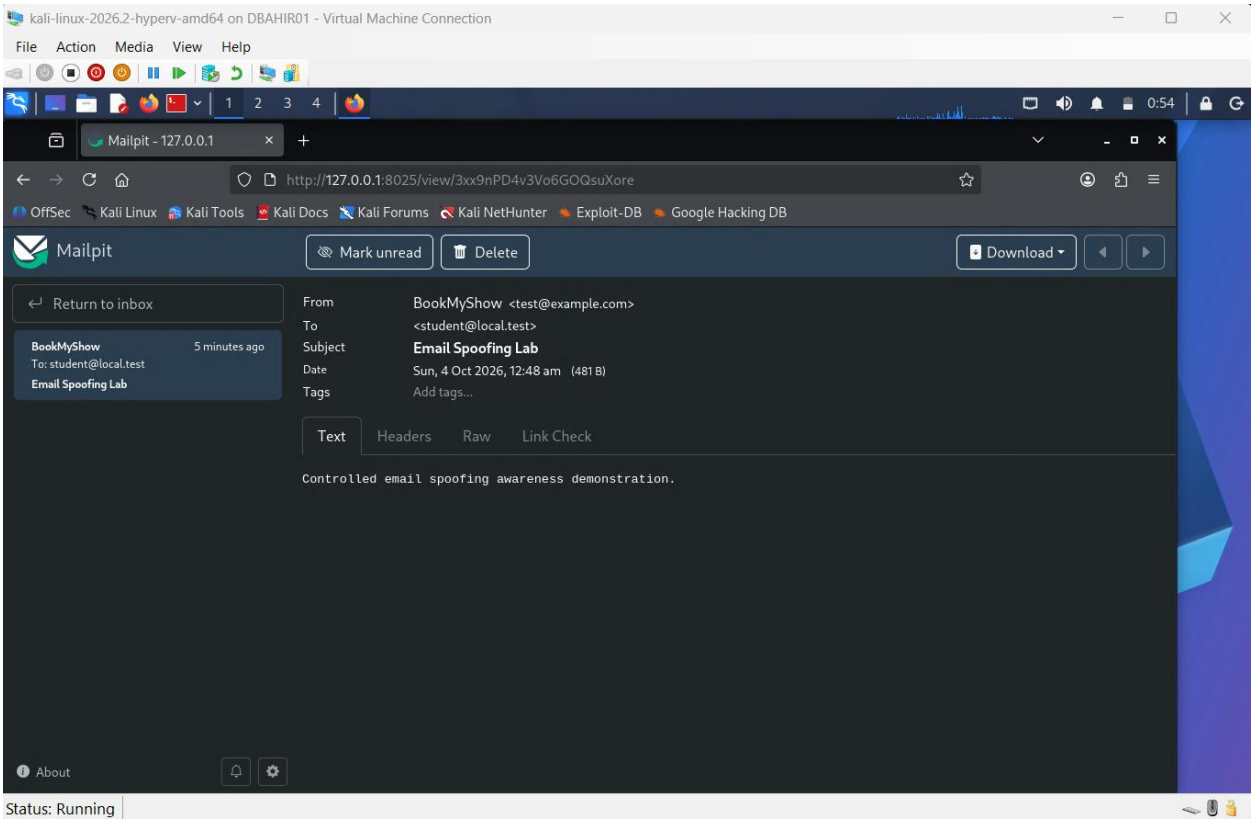
**3.Demonstrate email spoofing by sending a test email to yourself using a tool like 'sendmail' or an online SMTP testing service. Show how the sender's name can be faked to look like a message from BookMyShow or IRCTC.

Constraint: Do not use your real credentials or send emails to others — only test on your own email in a controlled setup.**

Ans :-




kali-linux-2026.2-hyperv-amd64 on DBAHIR01 - Virtual Machine Connection
File Action Media View Help
Mailpit - 127.0.0.1
http://127.0.0.1:8025
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Mailpit
Search mailbox
50
1-1 of 1
Inbox 1
Mark all read
Delete all
BookMyShow
To: student@local.test
Email Spoofing Lab
Controlled email spoofing awareness demonstration.
481 B
a few seconds ago
About
Status: Running


kali-linux-2026.2-hyperv-amd64 on DBAHIR01 - Virtual Machine Connection
File Action Media View Help
Mailpit - 127.0.0.1
http://127.0.0.1:8025/view/3xx9nPD4v3Vo6GOQsuXore
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Mailpit
Mark unread
Delete
Download
Return to inbox
BookMyShow 5 minutes ago
To: student@local.test
Email Spoofing Lab
From BookMyShow <test@example.com>
To <student@local.test>
Subject Email Spoofing Lab
Date Sun, 4 Oct 2026, 12:48 am (481 B)
Tags Add tags...
Text Headers Raw Link Check
Controlled email spoofing awareness demonstration.
About
Status: Running

4.Research and implement at least two prevention techniques against phishing or email spoofing (such as enabling 2FA or checking email headers) on your own email account. Take screenshots or write a short note explaining how you applied each technique.

Ans :-

❖ **Prevention Technique 1: Enable Two-Factor Authentication (2FA)**

To research and implement at least two security techniques to protect a personal email account from phishing attacks and email spoofing.

- Tools Required:

- Gmail account

- Internet connection

- Computer or smartphone

- Technique 1: Enable Two-Factor Authentication (2FA)
- Objective: Add an extra layer of security to the email account.

Procedure:

1. Open your Google Account: https://myaccount.google.com/security
2. Sign in to your Gmail account.
3. Navigate to Security → How you sign in to Google.
4. Select 2-Step Verification.
5. Follow the instructions to enable verification using a second method, such as Google Prompt or an authenticator app.
6. Confirm that 2-Step Verification is enabled.

❖ **Technique 2: Check Email Headers and Authentication**

Identify suspicious emails and detect possible sender spoofing.

- Procedure:

1. Open Gmail on your computer.

2. Open a received email (preferably a legitimate email from a known sender).

3. Click the three-dot menu (⋮) near the Reply button.

4. Select Show original.

5. Examine the email authentication results:

- SPF: PASS

- DKIM: PASS

- DMARC: PASS

6. Check whether the sender's email address matches the displayed sender name.

7. If an email appears suspicious, avoid clicking links or downloading attachments. Report it as phishing.

5.Explain how a phishing attack could target a Zomato or Swiggy user, and write a step-by-step guide for a new user on how to recognize and avoid such attacks. Include at least 3 warning signs to look for in fake emails or websites.

Ans :-

❖ How a Phishing Attack Could Target a Zomato/Swiggy User

Zomato and Swiggy are two of India's most widely used food delivery platforms, with hundreds of millions of registered users. Attackers exploit this massive user base because:

- **Frequent transactions** — Users order often, so a fake "order update" or "refund" message blends in naturally.
- **Sensitive data stored** — Saved addresses, phone numbers, email IDs, and linked payment methods (cards, UPI, wallets) make these accounts valuable.
- **Trust in the brand** — Users readily click links that appear to come from Zomato or Swiggy.
- **Urgency-driven communication** — Real order updates, delivery alerts, and refund notifications create a context where users act quickly without scrutinizing messages.

❖ How to Recognize and Avoid Phishing

- Step 1: Slow Down Before You Click
- Step 2: Verify the Sender's Email Address (Not Just the Name)
- Step 3: Inspect Links Before Clicking
- Step 4: Never Share OTPs, Passwords, or PINs
- Step 5: Go Directly to the App Instead of Clicking Links
- Step 6: Verify Through Official Customer Support
- Step 7: Secure Your Account
- Step 8: Report and Delete

- **In Gmail:** Three dots → Report phishing.
- **In Outlook:** Right-click → Report → Phishing.
- **In the Zomato/Swiggy app:** Use the "Report" or "Help" option to flag suspicious communication.
- **In India:** Report cyber fraud at cybercrime.gov.in or call **1930** (National Cyber Crime Helpline).

❖ Warning Signs of Fake Emails or Websites :-

1. Suspicious Website URL

- Fake websites may use misspelled domain names.
- Example: `zomato-offer.example.com` instead of the official Zomato domain.
- Always check the complete domain name before entering information.

2. Urgent or Attractive Offers

- Messages claiming "You won a free meal!"
- "Claim your ₹1,000 reward within 5 minutes."
- Attackers use urgency and attractive offers to make users act without checking.

3. Requests for OTP or Password

- Fake websites may ask for OTPs, passwords, or card details.
- Never share an OTP with someone claiming to be customer support.
- Do not enter sensitive information on an unverified website.

4. Poor Grammar and Unprofessional Design

- Spelling mistakes and unusual language.
- Low-quality logos or inconsistent branding.
- Unusual formatting in emails or messages.

5. Unknown Sender or Suspicious Attachments

- Emails from unfamiliar or misleading addresses.
- Unexpected attachments or links.
- Sender display names can be forged, so check the actual email address.

❖ Step-by-Step Guide for New Users to Recognize and Avoid Phishing

Step 1: Verify the Sender

Check the sender's complete email address or phone number. Do not trust a message only because it displays the company name or logo.

Step 2: Check the Website URL

Before opening a link, inspect the domain carefully. Prefer opening the official application directly instead of using links received through SMS or email.

Step 3: Do Not Share OTPs

Never disclose OTPs, passwords, UPI PINs, or other authentication details to callers, messages, or suspicious websites.

Step 4: Avoid Suspicious Offers

If a discount or reward looks unusually attractive, verify it through the official app's offers section.

Step 5: Use Secure Payment Methods

Use trusted payment options and verify the recipient and amount before approving a transaction. Never approve an unexpected collect request.

Step 6: Enable Account Security

Use a strong, unique password and enable available two-step verification or other account security features.

Step 7: Report Suspicious Messages

Report phishing emails through Gmail's Report phishing option. Report suspicious activity through the official food delivery application's support channel.

Step 8: Take Immediate Action if Information Is Shared

If you accidentally share your credentials or payment information:

- Change your account password immediately.
- Contact your bank or payment provider if financial information is involved.
- Block unauthorized payment methods where appropriate.
- Report financial cyber fraud in India through the official helpline 1930